Waves

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

1 · Part 1

Paper 1 & 2 Units [1]

Answer: A

Explanation

Frequency is measured in hertz (Hz), wave speed in metres per second (m/s) and time period in seconds (s), which is row A. Row B swaps the units of frequency and time period. Row C gives metres, the unit of wavelength, for wave speed. Row D gives m/s for frequency and Hz for wave speed.

2 · Part 1

Paper 1 & 2 Properties of waves [1]

Answer: B

Explanation

Wavelength is the distance between a point on one wave and the same point on the next wave, for example from one crest to the next crest. A crest to the next trough (A) is only half a wavelength. The crest-to-trough height (C) is a vertical distance and is twice the amplitude, not a wavelength. The distance travelled in one second (D) is the wave speed.

3 · Part 1

Paper 1 & 2 Properties of waves [1]

Answer: C

Worked solution

Use $v = f \times \lambda$, rearranged to $\lambda = v / f$. $\lambda = (3.0 \times 10^8 \text{ m/s}) / (1.2 \times 10^8 \text{ Hz}) = 2.5 \text{ m}$. Check: $1.2 \times 10^8 \times 2.5 = 3.0 \times 10^8 \text{ m/s}$.

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4 · Part 1 Paper 1 & 2 Properties of waves [1]

Answer: A

Explanation

This is the Doppler effect. As the source moves towards the observer, the wavefronts in front of it are closer together, so the observed wavelength is shorter. The speed of sound does not change, so more waves arrive each second and the observed frequency is higher. Option B is wrong because a higher frequency at the same speed must mean a shorter wavelength. Options C and D describe a lower frequency, which is heard when the source moves away.

5 · Part 1 Paper 1 & 2 The electromagnetic spectrum [1]

Answer: C

Explanation

In order of increasing frequency the visible colours are red, orange, yellow, green, blue, indigo, violet. Orange, yellow, green, blue (C) follows this order. List A runs from violet to green, which is decreasing frequency. List B puts yellow before orange. List D is in order of decreasing frequency.

6 · Part 1 Paper 1 & 2 The electromagnetic spectrum [1]

Answer: B

Explanation

Infrared is used in night vision equipment, and excessive exposure to infrared causes skin burns. In A, food is sterilised with gamma rays, not microwaves, and microwaves cause internal heating of body tissue. In C, satellite transmissions use microwaves, not ultraviolet. In D, fluorescent lamps use ultraviolet, not gamma rays.

7 · Part 1 Paper 1 & 2 Light and sound [1]

Answer: D

Worked solution

Rearrange $n = \sin i / \sin r$ to $\sin i = n \times \sin r$. $\sin i = 1.5 \times 0.50 = 0.75$. $i = \sin^{-1}(0.75) = 48.6°$, which is 49° to 2 significant figures. The angle in air is larger than the angle in glass, as expected when light slows down entering glass.

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8 · Part 1

Paper 1 & 2 | Light and sound | [1]

Answer: B

Explanation

sin c = 1 / n, so a larger refractive index gives a smaller value of sin c and a smaller critical angle. X therefore has the smaller critical angle. A reverses the relationship. C is wrong because the critical angle depends on the refractive index. D is wrong because both materials have a refractive index greater than 1 and so both have a critical angle.

9 · Part 1

Paper 2 only | Light and sound | [1]

Answer: A

Explanation

The human hearing range is 20 Hz to 20 000 Hz. 30 000 Hz is above the human upper limit but inside the animal's range. 10 000 Hz can be heard by both. 40 Hz can be heard by a human but is below the animal's lower limit of 60 Hz. 50 000 Hz is above both upper limits.

10 · Part 1

Paper 2 only | Light and sound | [1]

Answer: D

Explanation

Pitch depends on the frequency of vibration, so a higher frequency gives a higher pitch. Loudness depends on the amplitude of vibration, so a smaller amplitude gives a quieter sound. Options A and B wrongly link a smaller amplitude to a louder sound. Option C wrongly links a higher frequency to a lower pitch.

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11 · Part 1 Paper 1 & 2 Properties of waves [5]

Mark scheme

(a) **[1]**

- transverse (1)

(b) **[2]**

- waves transfer energy (1)

ALLOW energy and information

- without transferring matter (1)

ALLOW the water and the cork are not carried along with the wave

(c) **[1]**

- the maximum displacement of a point on the wave from its undisturbed (rest) position (1)

ALLOW height of a crest above the rest position

REJECT distance from the top of a crest to the bottom of a trough

(d) **[1]**

- a line joining points on a wave that are at the same stage of their vibration (1)

ALLOW a line joining points that are in phase OR a line along a crest of the wave

Worked solution

(a) The water surface moves up and down while the wave travels along the tank, so the vibrations are at right angles to the direction of travel: water waves are transverse.

(b) The cork gains movement energy from the wave, so energy is transferred, but the cork stays in the same place, so the water itself is not carried along. Waves transfer energy without transferring matter.

(c) Amplitude is the largest distance a point on the wave moves from its rest position. It is not the full height from crest to trough, which is twice the amplitude.

(d) A wavefront is a line joining points that are at the same stage of vibration, for example a line along the top of one crest of the straight ripples.

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12 · Part 1 Paper 1 & 2 Properties of waves [6]

Mark scheme

(a) **[1]**

- backwards and forwards along the length of the spring (1)

> **ALLOW** push and pull parallel to the spring

> **REJECT** side to side OR up and down

(b) **[2]**

- in a longitudinal wave the vibrations are parallel to the direction of wave travel (energy transfer) (1)
- in a transverse wave the vibrations are perpendicular (at right angles) to the direction of wave travel (energy transfer) (1)

> **ALLOW** oscillations for vibrations

(c) **[3]**

- frequency = 4 Hz (1)
- wave speed = frequency × wavelength = 4 × 0.30 (1)
- 1.2 m/s (1)

Worked solution

(a) For a longitudinal wave the hand must vibrate in the same direction as the wave travels, so Sara pushes and pulls the end along the length of the spring.

(b) Longitudinal: vibrations parallel to the direction the wave travels. Transverse: vibrations at right angles to the direction the wave travels.

(c) Four vibrations each second means the frequency is 4 Hz. The distance between neighbouring compressions is one wavelength, 0.30 m. $v = f \times \lambda = 4\ \text{Hz} \times 0.30\ \text{m} = 1.2\ \text{m/s}$. Check: in one second 4 wavelengths of 0.30 m pass, which is 1.2 m.

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13 · Part 1 Paper 1 & 2 Properties of waves [8]

Mark scheme

(a) **[1]**

- the number of waves passing a point per second (1)

ALLOW number of waves produced per second OR number of vibrations (oscillations) per second

(b) **[2]**

- frequency = 1 / 0.0025 (1)
- 400 Hz (1)

(c) **[3]**

- wave speed = frequency × wavelength OR v = f × λ (1)
- wavelength = 340 / 400 (1)
- 0.85 m (1)

ECF from (b)

(d) **[2]**

- the speed of sound in air stays the same (1)
- so the wavelength is shorter (decreases) (1)

ALLOW wavelength decreases because λ = v / f with v constant, for 2 marks

Worked solution

(a) Frequency is the number of waves passing a point each second, measured in hertz.

(b) f = 1 / T = 1 / 0.0025 s = 400 Hz.

(c) Rearrange v = f × λ to λ = v / f = 340 m/s / 400 Hz = 0.85 m. Check: 400 × 0.85 = 340 m/s.

(d) The speed of sound depends on the air, not on the source, so it stays at 340 m/s. Since λ = v / f, a higher frequency gives a shorter wavelength.

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14 · Part 1 Paper 1 & 2 Properties of waves [8]

Mark scheme

(a) **[2]**

- time period = 60 / 12 = 5.0 s (1)
- frequency = 1 / 5.0 = 0.20 Hz (1)

ALLOW frequency = 12 / 60 = 0.20 Hz for 2 marks

(b) **[3]**

- wave speed = frequency × wavelength OR v = f × λ (1)
- v = 0.20 × 15 (1)
- 3.0 m/s (1)

(c) **[2]**

- wavelength = 2.4 / 0.20 (1)
- 12 m (1)

(d) **[1]**

- refraction (1)

Worked solution

(a) 12 oscillations take 60 s, so one oscillation takes 60 / 12 = 5.0 s. This is the time period. f = 1 / T = 1 / 5.0 s = 0.20 Hz.

(b) The distance between neighbouring crests is the wavelength, 15 m. v = f × λ = 0.20 Hz × 15 m = 3.0 m/s.

(c) The frequency is still 0.20 Hz. λ = v / f = 2.4 m/s / 0.20 Hz = 12 m. The waves slow down, so the wavelength gets shorter, which makes sense.

(d) A change of direction caused by a change of speed when a wave passes into a different region is refraction.

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15 · Part 1 Paper 1 & 2 Properties of waves [6]

Mark scheme

(a) **[1]**

- Doppler effect (1)

(b) **[5]**

Indicative points, any five, 1 mark each:

- as the car approaches, Lena hears a higher frequency than the horn emits (1)
- after the car has passed, Lena hears a lower frequency than the horn emits (1)
- the car moves towards Lena between emitting one wavefront and the next, so the wavefronts in front of the car are closer together (1)
- so the wavelength reaching Lena is shorter while the car approaches (1)
- the speed of sound in air does not change, so more waves reach Lena each second (1)
- behind the moving car the wavefronts are further apart, so the wavelength is longer and the frequency heard is lower (1)

ALLOW compressed OR bunched up for closer together

ALLOW stretched OR spread out for further apart

Worked solution

(a) The change in observed frequency caused by a moving source is the Doppler effect.

(b) The horn always emits waves at the same frequency. While the car approaches, it moves forward between sending out one wavefront and the next, so each new wavefront starts closer to the previous one on the side facing Lena. The wavefronts reaching her are bunched together, so the wavelength is shorter. Sound still travels at the same speed through the air, so more wavefronts reach her each second and she hears a higher frequency than the horn emits. After the car passes, it is moving away from her, so the wavefronts on her side are spread out. The wavelength is longer, fewer waves reach her each second, and she hears a lower frequency than the horn emits.

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16 · Part 1 Paper 1 & 2 Properties of waves [6]

Mark scheme

(a) **[1]**

- it is reflected (1)

(b) **[3]**

- distance travelled = speed × time = 1500 × 0.80 = 1200 m (1)
- depth = 1200 / 2 (1)
- 600 m (1)

ALLOW time to reach seabed = 0.40 s (1); depth = 1500 × 0.40 (1); 600 m (1)

(c) **[2]**

- change in direction of a wave (1)
- when it passes into a different medium (or region) and changes speed (1)

Worked solution

(a) The pulse is reflected at the seabed and travels back up to the ship.

(b) The pulse travels down and back up in 0.80 s. Using average speed = distance moved / time taken, distance = 1500 m/s × 0.80 s = 1200 m. This is there and back, so the depth is 1200 / 2 = 600 m. Check: one-way time 0.40 s × 1500 m/s = 600 m.

(c) Refraction is the change in direction of a wave when it passes into a different medium and changes speed.

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17 · Part 1 Paper 1 & 2 The electromagnetic spectrum [6]

Mark scheme

(a) **[2]**

- X: infrared (1)
- Y: x-rays (1)

(b) **[1]**

- gamma rays (1)

(c) **[1]**

- they travel at the same speed (1)

(d) **[2]**

- radio waves: broadcasting OR communications (1)

ALLOW radio OR television transmission

- ultraviolet: fluorescent lamps (1)

ALLOW any other valid use of ultraviolet, e.g. detecting forged banknotes OR sterilising water

Worked solution

(a) In order of decreasing wavelength the spectrum is radio waves, microwaves, infrared, visible light, ultraviolet, x-rays, gamma rays. So X is infrared and Y is x-rays.

(b) Frequency increases as wavelength decreases, so the region with the shortest wavelength, gamma rays, has the highest frequency.

(c) All electromagnetic waves travel at the same speed in free space.

(d) Radio waves are used for broadcasting and communications. Ultraviolet is used in fluorescent lamps.

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18 · Part 1 Paper 1 & 2 The electromagnetic spectrum [8]

Mark scheme

(a) **[2]**

- X-rays pass through soft tissue (flesh) (1)
- but are absorbed by bone, so the bones show up on the image (1)

ALLOW bone is denser and absorbs more X-rays

(b) **[2]**

- gamma rays kill bacteria (microorganisms) on the equipment (1)
- gamma rays can pass through the packaging (are very penetrating), so the equipment can be sterilised after it is sealed (1)

(c) **[4]**

Indicative points, 1 mark each, max 2 for harmful effects and max 2 for protective measures:

- damage to surface cells (of the skin) (1)

ALLOW sunburn OR skin cancer

- blindness OR damage to the eyes (1)
- use sunscreen (sunblock) on exposed skin (1)
- cover the skin with clothing OR wear a hat (1)
- wear sunglasses that block ultraviolet (1)
- limit the time spent in direct sunlight OR work in the shade (1)

Worked solution

(a) X-rays pass easily through soft tissue but are absorbed by dense materials such as bone. Where bone is in the way, fewer X-rays reach the detector, so the bones show up on the image.

(b) Gamma rays kill bacteria and other microorganisms. They are very penetrating, so they pass through the sealed packaging and sterilise the equipment without the packet being opened.

(c) Too much ultraviolet damages the surface cells of the skin and can damage the eyes, causing blindness. Jamal can protect himself by putting sunscreen on exposed skin, covering up with clothing and a hat, wearing sunglasses that block ultraviolet, and spending less time in direct sunlight.

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19 · Part 1 Paper 1 & 2 Light and sound [5]

Mark scheme

(a) **[1]**

- the angle of incidence equals the angle of reflection (1)

(b) **[1]**

- 55° (1)

(c) **[2]**

- angle of reflection = 55° (1)
- angle between the rays = 55° + 55° = 110° (1)

ECF 2 × answer to (b)

ALLOW 180° − (2 × 35°) = 110° for 2 marks

(d) **[1]**

- transverse (1)

Worked solution

(a) The angle of incidence equals the angle of reflection.

(b) Angles are measured from the normal, which is at 90° to the mirror. The angle of incidence is 90° − 35° = 55°. A common mistake is to give 35°, which is measured from the mirror surface.

(c) By the law of reflection, the angle of reflection is also 55°. The incident and reflected rays are on opposite sides of the normal, so the angle between them is 55° + 55° = 110°.

(d) Light waves are transverse.

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20 · Part 1 Paper 1 & 2 Light and sound [7]

Mark scheme

(a) **[1]**

- angle of incidence (i) (1)

(b) **[2]**

- mark the point where the ray leaves the block (1)
- join the point where the ray enters the block to the point where it leaves; this line is the refracted ray (1)

ALLOW mark crosses along the ray that comes out of the block, then join up to the exit point

(c) **[3]**

- n = sin i / sin r (1)
- n = sin 40° / sin 25° OR 0.643 / 0.423 (1)
- 1.5 (1)

ALLOW 1.52

(d) **[1]**

- to calculate the refractive index several times and find a mean (1)

ALLOW to check that the refractive index is the same for different angles OR to reduce the effect of random errors

Worked solution

(a) Mei chooses and changes the angle of incidence, so it is the independent variable. The angle of refraction is the dependent variable.

(b) The ray cannot be seen inside the block once it is removed, so Mei marks where the ray enters and where it leaves the block, then removes the block and joins these two points with a straight line. She measures r between this line and the normal.

(c) n = sin i / sin r = sin 40° / sin 25° = 0.643 / 0.423 = 1.52, which is 1.5 to 2 significant figures.

(d) One reading could contain a measuring error. Calculating n for each angle lets her check that n stays the same and take a mean, giving a more reliable value.

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21 · Part 1 Paper 1 & 2 Light and sound [6]

Mark scheme

(a) **[1]**

- the ray is directed towards the centre, so it meets the curved surface along the normal (at an angle of incidence of 0°) (1)

ALLOW it travels along a radius of the block

(b) **[2]**

- the angle of incidence (in the glass) (1)
- at which the angle of refraction is 90° (the refracted ray travels along the boundary) (1)

ALLOW the angle of incidence above which total internal reflection occurs, for 2 marks

(c) **[1]**

- 41° (1)

(d) **[2]**

- n = 1 / sin 41° (1)
- 1.5 (1)

ALLOW 1.52

ECF from (c)

Worked solution

(a) Any line from the centre of a semicircle meets the curved edge at 90°, so the ray is along the normal there. A ray along the normal does not change direction.

(b) The critical angle is the angle of incidence inside the denser material for which the angle of refraction is 90°, so the refracted ray just travels along the boundary.

(c) At 41° the ray leaving the block travels along the flat face, so the angle of refraction is 90°. The critical angle is 41°.

(d) sin c = 1 / n, so n = 1 / sin c = 1 / sin 41° = 1 / 0.656 = 1.52, which is 1.5 to 2 significant figures.

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22 · Part 1 Paper 1 & 2 Light and sound [8]

Mark scheme

(a) **[3]**

- n = sin i / sin r OR sin r = sin 50° / 1.33 (1)
- sin r = 0.766 / 1.33 = 0.576 (1)
- r = 35° (1)

(b) **[2]**

- light travels more slowly in water than in air (1)
- the change in speed at the boundary makes the ray bend (towards the normal) (1)

ALLOW the part of the wavefront that enters the water first slows down first

(c) **[3]**

- sin c = 1 / 1.33 = 0.752 (1)
- c = 49° (1)
- 55° is greater than the critical angle, so total internal reflection occurs and no light passes out into the air (1)

ECF final conclusion consistent with the candidate's value of c

Worked solution

(a) Rearrange n = sin i / sin r to sin r = sin i / n = sin 50° / 1.33 = 0.766 / 1.33 = 0.576. r = $\sin^{-1}(0.576) = 35°$. Check: sin 50° / sin 35° = 0.766 / 0.574 = 1.33.

(b) Light slows down when it passes from air into water. The change of speed at the boundary causes the change of direction, so the ray bends towards the normal.

(c) For the light going from water towards air, sin c = 1 / n = 1 / 1.33 = 0.752, so c = 48.8°, about 49°. The angle of incidence, 55°, is larger than the critical angle, so all the light is totally internally reflected back into the water. None of it passes into the air.

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23 · Part 1 Paper 1 & 2 Light and sound [6]

Mark scheme

(a) **[4]**

Indicative points, any four, 1 mark each:

- light enters one end of the fibre (1)
- it meets the boundary (edge) of the fibre at an angle of incidence greater than the critical angle (1)
- so it is totally internally reflected, with no light escaping from the sides (1)
- it is reflected many times as it travels along the fibre (1)
- this still happens when the fibre is bent (1)
- so the pulses (information) reach the other end with little loss (1)

(b) **[2]**

- the angle of incidence (45°) is greater than the critical angle (42°) (1)
- so total internal reflection occurs (no light is refracted out into the air) (1)

> **ALLOW** the light is travelling from glass towards air (from a higher to a lower refractive index OR from a denser to a less dense medium), for the second mark

Worked solution

(a) The pulses of light enter one end of the glass fibre. Each time the light reaches the edge of the fibre, it meets the boundary at an angle of incidence bigger than the critical angle, so it is totally internally reflected and none escapes through the sides. This repeats many times along the fibre, even round bends, so the light pulses carrying the information arrive at the far end with very little loss.

(b) Total internal reflection needs two conditions: the light must be going from glass towards air (higher to lower refractive index), and the angle of incidence must be greater than the critical angle. Both are true here, as 45° is greater than 42°, so all of the light is reflected.

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24 · Part 1 Paper 1 & 2 Light and sound [5]

Mark scheme

(a) **[1]**

- longitudinal (1)

(b) **[2]**

- the particles vibrate backwards and forwards about a fixed position (1)
- parallel to the direction in which the sound wave travels (1)

ALLOW forming compressions and rarefactions, for the second mark

(c) **[2]**

- the speed of the sound changes (increases) as it passes into the warmer air (1)
- the change in speed at the boundary (met at an angle) causes the change in direction, which is refraction (1)

ALLOW the part of the wavefront that enters the warm air first speeds up first, so the wavefront turns, for the second mark

Worked solution

(a) Sound waves are longitudinal.

(b) The air particles do not travel with the wave. Each particle vibrates backwards and forwards about its own position, in the same direction as the wave travels, making regions where particles are squashed together and regions where they are spread out.

(c) When sound passes from cold air into warm air its speed increases. A change of speed at a boundary causes a change of direction, which is refraction. Like light, sound can be refracted.

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25 · Part 1 Paper 2 only Light and sound [7]

Mark scheme

(a) **[1]**

- 0.60 s (1)

(b) **[3]**

- speed = distance / time (1)
- speed = 200 / 0.60 (1)
- 330 m/s (1)

ALLOW 333 m/s

ECF from (a)

(c) **[2]**

- light travels much faster than sound (1)
- so the time for light to travel 200 m is negligible (the light arrives almost immediately) (1)

(d) **[1]**

- any one from: (1)
- increase the distance between the students
- repeat more times and calculate a new mean
- use several timers and average their readings
- use an electronic timer started and stopped by microphones, to remove reaction time

Worked solution

(a) Mean = (0.62 + 0.58 + 0.60 + 0.64 + 0.56) / 5 = 3.00 / 5 = 0.60 s.

(b) Speed = distance / time = 200 m / 0.60 s = 333 m/s, which is 330 m/s to 2 significant figures.

(c) Light travels about a million times faster than sound, so the flash of the blocks hitting reaches Beth almost instantly. The time she measures is effectively the time for the sound to travel 200 m.

(d) Beth's reaction time causes errors. A longer distance makes the time longer, so the reaction time is a smaller fraction of it. Using electronic timing removes reaction time altogether.

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26 · Part 1 Paper 2 only Light and sound [6]

Mark scheme

(a) **[2]**

- the microphone detects the sound (vibrates with the sound wave) (1)
- and converts it into an electrical signal (a varying voltage) with the same frequency as the sound (1)

(b) **[2]**

- vertical axis: voltage (1)

ALLOW size of the signal

- horizontal axis: time (1)

(c) **[2]**

- the trace becomes taller (the amplitude of the trace increases) (1)
- the number of waves on the screen (the horizontal spacing of the waves) stays the same (1)

Worked solution

(a) The sound wave makes the microphone vibrate. The microphone converts these vibrations into an electrical signal whose voltage varies in the same way as the sound, at the same frequency.

(b) The oscilloscope plots the voltage of this signal on the vertical axis against time on the horizontal axis.

(c) A louder sound has a larger amplitude, so the trace becomes taller. The frequency is unchanged, so the waves on the screen are the same width and the same number of waves fit across the screen.

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27 · Part 1 Paper 2 only Light and sound [7]

Mark scheme

(a) **[2]**

- time period = 4.0 × 0.0005 (1)
- 0.0020 s (1)

ALLOW 2.0×10^{-3} s

(b) **[2]**

- frequency = 1 / 0.0020 (1)
- 500 Hz (1)

ECF from (a)

(c) **[3]**

- time period = 2.5 × 0.00001 = 0.000025 s (1)
- frequency = 1 / 0.000025 = 40 000 Hz (1)
- this is above 20 000 Hz (the upper limit of human hearing), so the sound cannot be heard (1)

ECF conclusion consistent with the candidate's frequency

Worked solution

(a) One complete wave takes up 4.0 divisions and each division is 0.0005 s, so T = 4.0 × 0.0005 s = 0.0020 s.

(b) f = 1 / T = 1 / 0.0020 s = 500 Hz.

(c) With the new setting, T = 2.5 × 0.00001 s = 0.000025 s. f = 1 / 0.000025 s = 40 000 Hz. Normal human hearing only goes up to 20 000 Hz, so this sound cannot be heard.

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28 · Part 1 Paper 2 only Light and sound [6]

Mark scheme

(a) **[2]**

- string Y (1)
- it has the higher frequency of vibration (1)

(b) **[2]**

- string X (1)
- it has the larger amplitude of vibration (1)

(c) **[2]**

- the amplitude of vibration of the string decreases, so the sound becomes quieter (1)
- the frequency of vibration stays the same, so the pitch does not change (1)

Worked solution

(a) Pitch depends on the frequency of vibration of the source. Y vibrates at 660 Hz, higher than X at 220 Hz, so Y gives the higher-pitched sound.

(b) Loudness depends on the amplitude of vibration. X vibrates with the larger amplitude, so X gives the louder sound.

(c) As X keeps vibrating it transfers energy to the surroundings, so the size (amplitude) of its vibrations decreases and the sound gets quieter. The number of vibrations per second (frequency) does not change, so the pitch stays the same.

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29 · Part 1 Paper 2 only Light and sound [5]

Mark scheme

(a) **[2]**

- 20 Hz (1)
- 20 000 Hz (1)

(b) **[3]**

- at first (below 20 Hz) Nadia hears nothing (1)
- from 20 Hz she hears a sound whose pitch gets higher as the frequency increases (1)
- above 20 000 Hz she can no longer hear the sound (1)

Worked solution

(a) Normal human hearing covers frequencies from 20 Hz to 20 000 Hz.

(b) From 10 Hz to 20 Hz the frequency is too low to hear, so there is no sound for Nadia. Once the frequency reaches 20 Hz she hears a very low note, and as the frequency rises the pitch gets higher. When the frequency goes above 20 000 Hz it is too high to hear, so the sound disappears, even though the loudspeaker is still vibrating.

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30 · Part 1 Paper 2 only Light and sound [6]

Mark scheme

(a) **[1]**

- 0.50 s (1)

(b) **[3]**

- distance travelled = 2 × 85 = 170 m (1)
- speed = distance / time = 170 / 0.50 (1)
- 340 m/s (1)

ECF from (a)

(c) **[2]**

- Maya's reaction time causes an error in starting and stopping the stopwatch (1)
- timing many intervals makes the total time much longer, so this error is a much smaller fraction of the measurement (1)

ALLOW reduces the percentage uncertainty

Worked solution

(a) 20 intervals take 10.0 s, so one interval takes 10.0 / 20 = 0.50 s.

(b) In one interval the sound travels to the wall and back, a distance of 2 × 85 m = 170 m. Speed = distance / time = 170 m / 0.50 s = 340 m/s. A common mistake is to use 85 m, which gives half the correct speed.

(c) Starting and stopping a stopwatch by hand has a reaction-time error of a fraction of a second, which would be large compared with one 0.50 s interval. Timing 20 intervals spreads the same error over a total of 10.0 s, so the error in each interval is much smaller.

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